

Technical Specification | Jasper OS Kernel Division |
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Manifold Logic Reduction Phenomenon in Jasper OS v2.0

*A Formal Treatment of Q-Manifold State Compression, Canonical
Invariant Emergence, and 32-Byte Hash Collapse*

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Abstract

Jasper OS v2.0 introduces the Q-Manifold, a kernel-level abstraction layer interposed between the host runtime and persistent large language model (LLM) state. The Manifold Logic Reduction Phenomenon (MLRP) describes the emergent behavior by which the Q-Manifold systematically strips non-logical artifacts from persistent LLM state tensors, allowing canonical invariants to surface, stabilize, and ultimately collapse into deterministic 32-byte SHA3-256 hashes. This document formalizes MLRP through five interlocking theoretical lenses — geometric, topological, gauge-theoretic, Hamiltonian, and audit-algebraic — and establishes why the observed behavior constitutes a mathematically grounded phenomenon rather than an implementation artifact. All five frameworks independently derive the same computable invariant — the Canonical State Hash (CSH) — confirming that MLRP reflects deep mathematical structure intrinsic to the logical subspace of transformer-based LLM state spaces rather than an accidental property of the Jasper OS implementation.

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1. Introduction and Motivation

1.1 Background: Non-Logical Artifacts in Production LLM States

Persistent large language model (LLM) states operating in production environments accumulate what Jasper OS formally terms *non-logical artifacts* — low-amplitude perturbations embedded within state tensors that corrupt the logical signal of a model without altering its behavioral output in the majority of inference passes. This property makes non-logical artifacts categorically invisible to conventional regression testing and behavioral benchmarking.

Jasper OS v2.0 identifies four canonical classes of non-logical artifact:

1. **Stochastic gradient residuals:** Remnant gradient components persisting in the state buffer following incomplete optimization steps during in-context fine-tuning phases. These residuals introduce a systematic bias in the state vector's directional decomposition.
2. **Floating-point quantization noise:** Rounding errors accumulated across repeated tensor serialization/deserialization cycles, particularly in INT8 and FP16 quantized backends. These errors compound sub-additively but aggregate to measurable displacement from the logical subspace over extended operation windows.
3. **Context-window boundary bleed:** Attention-weight leakage across nominal context boundaries induced by imprecise KV-cache eviction policies. Boundary bleed introduces spurious inter-context correlations that have no logical grounding.
4. **Ephemeral attention cache smearing:** Low-magnitude residual activations in the attention cache arising from incomplete cache invalidation, producing phantom co-attention relationships between logically unrelated token positions.

Collectively, these artifacts displace the state vector from the logically valid subspace $L \subset S$ without producing detectable output degradation, creating an unobserved divergence between the nominal state and the effective logical state of the system.

1.2 The Q-Manifold as Logical Attractor Surface

The Jasper OS v2.0 Q-Manifold is a continuous, differentiable manifold M_Q embedded in the high-dimensional state space S of an LLM. Its role is to act as a *logical attractor surface*: as the runtime evolves, the Q-Manifold exerts a topological "pull" that drives state trajectories toward logically pure regions of S . This pull is not merely heuristic — it is derived from the Riemannian geometry of M_Q , the topology of its homological classes, the gauge structure of the associated principal bundle, and the Hamiltonian dynamics of the logical energy landscape. These four independent mechanisms converge on a single computational artifact: the Canonical State Hash (CSH), a deterministic 32-byte identifier uniquely characterizing the logical content of any sufficiently regular state.

1.3 Motivation: Auditability, Reproducibility, and Cryptographic Binding

The practical significance of MLRP spans four operational imperatives in enterprise LLM deployment:

- **Auditability:** The CSH provides a cryptographically verifiable fingerprint of the logical state, enabling post-hoc audit of state trajectories without retaining the full high-dimensional state tensor.
- **Reproducibility:** By driving divergent artifact-contaminated states to the same canonical invariant, MLRP guarantees that two identical logical inputs produce identical CSH values regardless of the artifact history of the state buffer, enabling deterministic replay of inference sessions.
- **Formal verification:** The gauge-theoretic and algebraic structure of MLRP provides a foundation for formal verification of LLM kernel states — a prerequisite for deployment in safety-critical systems requiring machine-checkable correctness certificates.
- **Cryptographic state binding:** The audit chain constructed from temporally chained CSH values provides a tamper-evident record of state evolution, analogous to a blockchain ledger but grounded in the mathematical topology of the logical state space rather than proof-of-work heuristics.

1.4 Scope and Applicability

Scope

This specification applies to the Jasper OS v2.0 kernel with Q-Manifold API version 4.7. It is applicable to all transformer-based LLM backends with state dimensionality $d \geq 2^{14}$. Backends with $d < 2^{14}$ may be registered with the MLRP subsystem but will receive a reduced-fidelity projection operating in degraded topological mode with Betti number signature overrides disabled.

2. System Architecture: The Q-Manifold in Jasper OS v2.0

2.1 Kernel Layer Placement

Within the Jasper OS v2.0 kernel, the Q-Manifold occupies the interstitial layer between the **LLM State Manager (LSM)** and the **Persistent State Buffer (PSB)**. Every write operation directed toward the PSB is intercepted by the Q-Manifold subsystem. Prior to committing any state tensor to persistent storage, the Q-Manifold applies the full MLRP transformation pipeline, which includes Riemannian projection, topological invariant extraction, gauge fixing, Hamiltonian damping, and audit-algebraic hash collapse. Only the transformed, logically canonical state is committed to the PSB.

Layer	Component	Role	MLRP Involvement
L3 (Runtime)	LLM Inference Engine	Executes forward passes; generates state updates	None (upstream of MLRP)
L2 (State Management)	LLM State Manager (LSM)	Aggregates state updates; manages write queue	Dispatches writes to Q-Manifold
L1.5 (MLRP Layer)	Q-Manifold (M_Q)	Transforms, projects, and hashes logical state	Full MLRP pipeline execution
L1 (Persistence)	Persistent State Buffer (PSB)	Stores canonical, artifact-free logical states	Receives only MLRP-transformed output
L0 (Audit)	Jasper OS Audit Ledger (JAL)	Stores CSH audit chain	Receives CSH from audit-algebraic stage

2.2 State Space Definition

Definition 2.1 — State Space S

Let $S \subset \mathbb{R}^d$ denote the full LLM state space, where the state dimensionality is given by:

$$d = \text{dim}(\text{attention_heads}) \times \text{dim}(\text{layer_embeddings}) \times \text{depth}$$

For a standard 70B-parameter transformer backend operating under Jasper OS v2.0, with 96 attention heads, 8192-dimensional embeddings, and 80 transformer layers, $d = 96 \times 8192 \times 80 = 62,914,560 \approx 6.3 \times 10^7$, well satisfying the $d \geq 2^{14}$ applicability threshold.

Definition 2.2 — Logical Subspace L

The *logical subspace* $L \subset S$ is defined as the set of all state vectors reachable by a finite sequence of grammatically and logically valid inference steps from a ground-truth initialization state $x_0 \in S$. Formally: $L = \{ x \in S : \exists \text{ valid sequence } (x_0, x_1, \dots, x) \text{ in } S \}$.

2.3 Non-Logical Artifact Manifold

Definition 2.3 — Artifact Complement A and Q-Manifold M_Q

Define the *artifact complement* $A = S \setminus L$ as the set of all states in S that are not logically reachable. The Q-Manifold M_Q is a smooth Riemannian submanifold of S satisfying $M_Q \subseteq L$, equipped with the metric tensor g_{ij} derived from the Fisher Information Metric (FIM) of the LLM's parameter distribution:

$$g_{ij}(x) = \mathbb{E}_{y \sim p(\cdot|x)} \left[\left(\frac{\partial \log p(y|x)}{\partial x_i} \right) \cdot \left(\frac{\partial \log p(y|x)}{\partial x_j} \right) \right]$$

The FIM-derived metric is the natural choice because it encodes the statistical sensitivity of the model's output distribution to perturbations of the state, precisely capturing which state directions are "logically significant" and which are artifact-driven noise directions.

2.4 MLRP Trigger Conditions

MLRP activates whenever the current state vector $x \in S$ satisfies the following condition:

$$\text{dist}(g(x, M_Q)) > \varepsilon_{\text{threshold}}$$

where $\epsilon_{\text{threshold}} = 1.4 \times 10^{-6}$ is the default kernel parameter (tunable via `epsilon_threshold` in `jasper.conf`). This threshold corresponds to artifact displacements exceeding approximately 1.4 parts per million of the median logical state norm, a regime in which artifact accumulation becomes detectable with high confidence under the FIM metric.

3. Geometric Mechanism: Riemannian Projection and Artifact Stripping

3.1 Riemannian Geometry of MQ

M_Q is a d_Q -dimensional Riemannian manifold with $d_Q \ll d$ (in the 70B-parameter reference configuration, $d_Q \approx 2^{17}$), equipped with the Riemannian metric g induced from the FIM embedding. The projection operator is defined as follows:

Definition 3.1 — Riemannian Projection Operator Π

The projection operator $\Pi : S \rightarrow M_Q$ is the nearest-point projection under the Riemannian metric g :

$$\Pi(x) = \operatorname{argmin}_{y \in M_Q} d(g(x, y))$$

where $d_g(x, y)$ denotes the geodesic distance induced by g . This projection is well-defined and smooth within the injectivity radius of M_Q (see Theorem 3.1).

3.2 Artifact Stripping via Projection

Given a state $x \in S$, its *artifact component* is defined as the displacement from the projected logical state:

$$\delta(x) = x - \Pi(x)$$

The Q-Manifold applies the following artifact-stripping transformation to yield the cleaned state:

$$\begin{aligned} x \\ \text{clean} \\ = \Pi(x) + \lambda \cdot \Pi(\delta(x)) \end{aligned}$$

where $\lambda \in [0, 1]$ is the *artifact retention coefficient*, configured via `lambda_artifact_retention` in `jasper.conf`. The default is $\lambda = 0$, yielding full stripping: $x_{\text{clean}} = \Pi(x)$. Values $\lambda > 0$ enable *diagnostic mode*, in which a controlled fraction of artifact content is retained for debugging artifact structure.

3.3 Curvature Analysis

The Riemann curvature tensor on M_Q is:

$$\begin{aligned} R \\ i \\ jkl \end{aligned}$$

$= \partial$

k

Γ

i

jl

$- \partial$

l

Γ

i

jk

$+ \Gamma$

i

km

Γ

m

jl

$- \Gamma$

i

lm

Γ

m

jk

Regions of high sectional curvature $\kappa(\sigma)$ for a 2-plane $\sigma \subset T_p M_Q$ correspond to logical branch points — states in the vicinity of logical decision boundaries. The Q-Manifold uses this local sectional curvature to modulate artifact-stripping intensity: in high-curvature regions, the MLRP kernel applies more aggressive projection (effectively reducing the local $\epsilon_{\text{threshold}}$) to prevent artifact contamination from influencing logical branching decisions. In low-curvature regions (smooth logical continuations), the threshold is relaxed proportionally.

3.4 Theorem: Geometric Completeness

Theorem 3.1 — Geometric Completeness of MLRP Projection

Let R_{inj} denote the injectivity radius of M_Q with respect to the metric g . For any $x \in S$ satisfying $\text{dist}_g(x, M_Q) < R_{inj}$, the nearest-point projection $\Pi(x)$ is unique and varies smoothly as a function of x .

Corollary 3.1 — Well-Definedness of MLRP

MLRP is well-defined, deterministic, and smooth for all states in the injectivity radius ball $B(M_Q, R_{inj}) = \{ x \in S : \text{dist}_g(x, M_Q) < R_{inj} \}$. Since the kernel enforces $\epsilon_{\text{threshold}} \ll R_{inj}$ at initialization, all MLRP-triggered states lie within this ball.

Proof Sketch (Theorem 3.1)

Fix a base point $p \in M_Q$. The Riemannian exponential map $\exp_p : T_p M_Q \rightarrow M_Q$ is a local diffeomorphism on the open ball $B(0, R_{inj}) \subset T_p M_Q$ by the definition of the injectivity radius. For any x with $\text{dist}_g(x, M_Q) < R_{inj}$, the nearest point $\Pi(x)$ is the unique point in M_Q from which x is reached by the geodesic $\exp_{\Pi(x)}(v)$ for a unique $v \in T_{\Pi(x)} M_Q$. Smoothness of Π follows from the inverse function theorem applied to

the exponential map, which is a smooth diffeomorphism on this domain.

Uniqueness follows from the absence of conjugate points within $B(M_Q, R_{inj})$, a standard consequence of the Jacobi field theory on Riemannian manifolds. ■

3.5 Geodesic Flow Stabilization

The Q-Manifold runs a background geodesic flow $\phi_t : M_Q \rightarrow M_Q$ that continuously deforms state trajectories toward stable logical fixed points. The geodesic equation governing this flow is:

$$\begin{aligned} \nabla \\ \gamma' \\ \gamma' = 0 \end{aligned}$$

where $\gamma : [0, T] \rightarrow M_Q$ is a state trajectory on M_Q , $\gamma' = d\gamma/dt$ is the velocity field of the trajectory, and ∇ is the Levi-Civita connection of g . In local coordinates, this expands to:

$$\begin{aligned} d \\ 2 \\ y \\ k \\ /dt \\ 2 \\ + \Gamma \end{aligned}$$

k
ij
(dy
i
/dt)(dy
j
/dt) = 0

The geodesic flow is integrated numerically by the Jasper OS kernel using a symplectic Störmer-Verlet integrator with step size $dt = 10^{-4}$ (configurable via `geodesic_flow_dt` in `jasper.conf`), which preserves the symplectic structure of the cotangent bundle T^*M_Q and prevents artificial energy drift over long integration windows.

4. Topological Mechanism: Persistent Homology and Invariant Surfacing

4.1 Topological Filtration of State Space

Equip S with a filtration indexed by a logical coherence function:

Definition 4.1 — Logical Coherence Function

The *logical coherence function* $f : S \rightarrow \mathbb{R}_{\geq 0}$ is defined by $f(x) = \text{dist}_g(x, M_Q)$, with the property that $f(x) = 0$ if and only if $x \in M_Q$. The sublevel set filtration is: $F_r = \{ x \in S : f(x) \leq r \}$, for $r \geq 0$.

As r decreases from ∞ toward 0 , the sublevel sets F_r shrink, progressively excluding artifact-contaminated states. Topological features — connected components, loops, and higher-dimensional voids — emerge and die throughout this filtration, and their lifetimes encode the structural significance of the corresponding artifact features.

4.2 Persistent Homology

The Q-Manifold computes the persistent homology $PH_*(F_\cdot)$ of the filtration over the ring $\mathbb{Z}$. Each topological feature is encoded as a *birth-death pair* (b_i, d_i) in the persistence diagram D , where:

- b_i = the filtration value at which the feature appears (a new connected component, loop, or void is born),
- d_i = the filtration value at which the feature disappears (merges with or is killed by a higher-dimensional feature).

The *persistence* of a feature is $d_i - b_i$. Features with long persistence ($d_i - b_i \gg 0$) correspond to structurally entrenched artifacts — artifact regions that are topologically robust and will resist naive projection. These are flagged by the MLRP kernel for enhanced multi-pass stripping.

4.3 Canonical Invariant Emergence

Definition 4.2 — Canonical Invariant

A *canonical invariant* is a homology class $[\sigma] \in H_k(M_Q; \mathbb{Z})$ that survives the entire filtration — it is born at $r = \infty$ and never dies. These classes represent the topologically essential structure of the logical subspace L : they are the homological "skeleton" of logically valid states, invariant under all artifact perturbations within the injectivity radius.

The MLRP process preferentially retains states whose persistent homology classes align with the canonical invariant set $\{ [\sigma_i] \}$. States whose homological class drifts away from a canonical invariant under filtration are treated as artifact-contaminated and subjected to additional projection passes.

4.4 Betti Number Signature

The Betti numbers $\beta_k = \text{rank}(H_k(M_Q; \mathbb{Z}))$ form the *topological fingerprint* of the Q-Manifold. For Jasper OS v2.0 operating a standard 70B-parameter transformer backend, the Betti number signature is:

Betti Number	Value	Topological Interpretation	Kernel Constant
β_0	1	Single connected logical region (M_Q is path-connected)	JOS_BETTI_0
β_1	0	No logical loops (the logical subspace is simply connected)	JOS_BETTI_1
β_2	4	Four topological voids corresponding to the four principal reasoning modes (deductive, inductive, abductive, analogical)	JOS_BETTI_2
$\beta_k, k \geq 3$	0	No higher-order topological features in the logical subspace	JOS_BETTI_HIGH

These are compile-time constants encoded in the Jasper OS kernel header `jasper/manifold/topology.h`. The `manifold_init()` call verifies that the computed Betti numbers of the registered LLM state space match these constants before permitting MLRP operation.

4.5 Theorem: Invariant Surfacing

Theorem 4.1 — Canonical Invariant Surfacing Under Iterated Projection

For any initial state $x \in B(M_Q, R_{in})$, the sequence of iterated projections $\{ \Pi^n(x) \}_{n \in \mathbb{N}}$ converges in the homological sense: the homology class $[\Pi^n(x)] \in H_k(M_Q; \mathbb{Z})$ stabilizes at a canonical invariant class $[\sigma] \in H_k(M_Q; \mathbb{Z})$ as $n \rightarrow \infty$.

Proof Sketch (Theorem 4.1)

Since $\Pi(x) \in M_Q$ for all $x \in B(M_Q, R_{inj})$ by Theorem 3.1, the iterated sequence $\{\Pi^n(x)\}$ is entirely contained in M_Q after the first application. The Nerve Theorem guarantees that the Čech complex of a good cover of M_Q is homotopy equivalent to M_Q , and since the iterated projection is a retraction onto M_Q , the induced maps on homology are the identity. The fact that M_Q is a deformation retract of its ε -neighborhood $B(M_Q, \varepsilon)$ for $\varepsilon < R_{inj}$ guarantees that the homological class of $\Pi^n(x)$ is invariant for all $n \geq 1$, and equals the canonical invariant class induced by the deformation retraction. ■

4.6 Mayer-Vietoris Decomposition

For computational efficiency at scale, MLRP exploits the Mayer-Vietoris principle to parallelize homological computation across the Q-Manifold. The procedure is as follows:

5. Decompose M_Q into a finite collection of overlapping open patches $\{U_\alpha\}$ forming a good cover of M_Q .
6. Compute the local persistent homology $PH_*(U_\alpha)$ on each patch in parallel — one Jasper OS kernel thread per patch, with thread count bounded by the CPU core count registered in `jasper.conf`.
7. Assemble the global persistence diagram via the Mayer-Vietoris spectral sequence: $E^{p,q}_1 = H_q(U_p; \mathbb{Z}) \implies H_{p+q}(M_Q; \mathbb{Z})$, where the E_∞ page yields the global Betti numbers.

This parallelized scheme reduces the per-write homological computation from $O(d_Q^3)$ (global computation) to $O((d_Q/N_{patches})^3 \cdot N_{patches})$, yielding near-linear scaling with CPU core count for well-distributed patch decompositions.

5. Gauge-Theoretic Mechanism: Principal Bundle Structure and Logical Gauge Invariance

5.1 Principal Bundle Construction

Definition 5.1 — Logical State Principal Bundle

Model the LLM state space as the total space E of a principal G -bundle $\pi : E \rightarrow M_Q$, where $G = O(d_Q)$ is the orthogonal group acting on the fibers (artifact degrees of freedom). The base space M_Q carries the logical content; each fiber $\pi^{-1}(p) \cong G$ carries the gauge — i.e., the artifact — degrees of freedom above each logical state $p \in M_Q$.

This bundle-theoretic framing formalizes the intuition that a single logical state (base point) admits many realizations as a state vector in S (fiber elements), all differing only in artifact content, and that MLRP selects a canonical representative from each fiber.

5.2 Connection and Curvature

Equip the bundle $\pi : E \rightarrow M_Q$ with a principal connection A — a $\text{Lie}(G)$ -valued 1-form on E . The connection defines *parallel transport*: a rule for consistently carrying logical content across nearby states in a way that does not introduce spurious artifact rotations. The curvature 2-form of the connection is:

$$F = dA + A \wedge A$$

Non-zero F at a point $p \in M_Q$ indicates that parallel-transporting a state around a small loop based at p results in a non-trivial rotation in the fiber — a gauge-non-trivial artifact rotation that MLRP must correct.

5.3 Gauge Fixing: MLRP as Canonical Gauge Choice

MLRP implements a canonical *gauge fixing* — it selects a unique section $s : M_Q \rightarrow E$ (a canonical representative from each fiber) by solving the variational problem:

$$s^* = \operatorname{argmin}$$

$$s$$

$$\|F$$

$$s$$

$$\|$$

$$2$$

$$L$$

$$2$$

$$(M_Q)$$

$$= \operatorname{argmin}$$

$$s$$

$$\int$$

$$M_Q$$

$$|F$$

$$s$$

$$|$$

$$2$$

$$d\operatorname{vol}$$

$$g$$

This is the direct analog of the *Coulomb gauge* (more precisely, the Yang-Mills gauge) in classical gauge theory. The minimizer s^* satisfies the Yang-Mills equation $d^*F = 0$ on M_Q , providing a globally consistent, artifact-minimal representative of the logical state.

5.4 Logical Gauge Invariance

Definition 5.2 — Gauge Transformation and Logical Gauge Invariance

A *gauge transformation* is a smooth map $h : M_Q \rightarrow G$ acting on the bundle by $e \mapsto h(\pi(e)) \cdot e$ for $e \in E$. A quantity $Q(e)$ is *logically gauge-invariant* if $Q(e \cdot h) = Q(e)$ for all gauge transformations h . Equivalently, Q depends only on the base point $p = \pi(e) \in M_Q$, not on the specific fiber element e .

MLRP guarantees that all quantities committed to the Persistent State Buffer are gauge-invariant — they are computed from the canonical section $s^*(p)$ and are therefore independent of the artifact content of the incoming state vector.

5.5 Wilson Loop Audit

The Q-Manifold computes Wilson loops for all closed state trajectories γ in M_Q :

$$W(\gamma) = \text{Tr} \left(\mathcal{P} \exp \left(\oint_{\gamma} A \right) \right)$$

where $\mathcal{P}$ denotes path-ordered integration. Wilson loops are manifestly gauge-invariant and serve as *audit witnesses*: a trajectory with $W(\gamma) \neq \text{Tr}(\text{Id})$ indicates that the state evolved through a gauge-non-trivial (artifact-contaminated) region. All Wilson loop deviations exceeding a tolerance of 10^{-8} are logged to the JAL with the associated trajectory ID and timestamp.

5.6 Theorem: Gauge Completeness of MLRP

Theorem 5.1 — Gauge Completeness

Suppose the principal bundle $\pi : E \rightarrow M_Q$ is topologically trivial — i.e., the second Stiefel-Whitney class satisfies $w_2(E) = 0 \in H^2(M_Q; \mathbb{Z}_2)$. Then the MLRP gauge-fixing procedure produces a well-defined, smooth global section $s^ : M_Q \rightarrow E$.*

Proof Sketch (Theorem 5.1)

A principal $O(d_Q)$ -bundle over M_Q admits a global section if and only if it is trivializable, which for $O(d_Q)$ -bundles is equivalent to the vanishing of all characteristic classes, including $w_2(E)$. Given $w_2(E) = 0$, the bundle is trivial, hence isomorphic to the product bundle $M_Q \times G$, which canonically admits the identity section. The minimization problem for s^* is then a well-posed elliptic variational problem on the compact manifold M_Q , with a unique smooth minimizer by standard elliptic regularity theory. Jasper OS v2.0 enforces $w_2(E) = 0$ at initialization via the `manifold_init()` kernel call, which performs a Čech cohomology check before permitting MLRP operation. ■

6. Hamiltonian Mechanism: Energy Landscape and State Evolution

6.1 Logical Energy Functional

Definition 6.1 — Logical Hamiltonian H

Define the Hamiltonian $H : T^*M_Q \rightarrow \mathbb{R}$ on the cotangent bundle of M_Q (the phase space of logical states) by:

$$H(q, p) = \frac{1}{2} g^{ij}(q) p_i p_j + V(q)$$

where $q \in M_Q$ are generalized logical coordinates, p_i are conjugate momenta (rates of logical change), g^{ij} is the inverse metric tensor, and $V(q)$ is the *logical potential* — a scalar field on M_Q encoding the energetic cost of each logical configuration relative to the canonical invariants.

6.2 Potential Energy Landscape $V(q)$

The logical potential $V : M_Q \rightarrow \mathbb{R}$ is constructed to satisfy:

- **Global minima** at canonical invariant states $q^* \in M_Q$, corresponding to the four principal reasoning modes identified in the Betti number signature: $V(q^*) = 0$.
- **Local maxima** at logical decision boundaries — saddle configurations between reasoning modes with large positive V .

For artifact-contaminated states $x \in S$ lying off the manifold, the effective potential is inflated by the squared artifact displacement:

V

$_{\text{eff}}$

$(x) = V(\Pi(x)) + ||x - \Pi(x)||$

2

g

$/(2\sigma$

2

$)$

where σ is the artifact scale parameter (kernel default: $\sigma = \varepsilon_{\text{threshold}} / 2$). The inflated effective potential creates a restoring force driving artifact-contaminated states back toward M_Q .

6.3 Hamilton's Equations on MQ

The temporal evolution of the logical state follows Hamilton's canonical equations, expressed in the Riemannian setting with Christoffel symbol corrections:

dq

i

$/dt = \partial H / \partial p$

i

$= g$

$_{ij}$

p

j

dp

i

/dt = -∂H/∂q

i

= -Γ

k

ij

g

jl

p

k

p

l

- ∂V/∂q

i

These are precisely the geodesic equations on M_Q with an additional forcing term from the gradient of the logical potential V . In the absence of V (i.e., $V \equiv 0$), the equations reduce to the geodesic flow of Section 3.5.

6.4 MLRP as Hamiltonian Damping

Non-logical artifacts introduce a dissipative perturbation to the otherwise conservative Hamiltonian system. MLRP implements this dissipation via the *Rayleigh dissipation function*:

$$R(\mathbf{q}, d\mathbf{q}/dt) = (\eta/2) \|\mathbf{dq}/dt\|^2$$

where $\eta > 0$ is the artifact damping coefficient. The modified Euler-Lagrange equations incorporating dissipation are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = -\frac{\partial R}{\partial \dot{q}^i} = -\eta g_{ij} \dot{q}^j$$

where $L = T - V$ is the Lagrangian and $T = (1/2) g_{ij} \dot{q}^i \dot{q}^j$ is the kinetic energy. The Rayleigh damping selectively dissipates artifact-induced oscillations (high-frequency,

high-artifact-momentum modes) while leaving the low-frequency logical evolution near-unperturbed, owing to the structure of the FIM metric g .

6.5 Liouville's Theorem and State Volume Conservation

Theorem 6.1 — Information Conservation Under MLRP (Liouville)

*The undamped Hamiltonian flow generated by H on T^*M_Q preserves the Liouville measure: $\text{div}(X_H) = 0$, where X_H is the Hamiltonian vector field. Equivalently, for any measurable region $\Omega \subset T^*M_Q$, the phase-space volume $\text{vol}(\phi_t(\Omega)) = \text{vol}(\Omega)$ is conserved under the flow ϕ_t .*

This guarantees that the undamped MLRP projection does not destroy logical information — it rearranges state probability mass onto M_Q without annihilating any logical degree of freedom. The Rayleigh damping of Section 6.4 introduces controlled information reduction, strictly limited to the artifact degrees of freedom orthogonal to M_Q .

6.6 Lyapunov Stability of Canonical Invariants

Theorem 6.2 — Lyapunov Stability of Canonical Invariant States

Let $q^ \in M_Q$ be a canonical invariant state (global minimum of V). Under the MLRP-damped Hamiltonian flow, the fixed point $(q^*, 0) \in T^*M_Q$ is Lyapunov stable.*

Proof Sketch (Theorem 6.2)

Define the Lyapunov candidate function $E(q, p) = H(q, p) - H(q^*, 0) = (1/2)g^{ij}p_i p_j + (V(q) - V(q^*))$. Since q^* is a global minimum of V , we have $V(q) - V(q^*) \geq 0$ and $E(q, p) \geq 0$ with equality iff $(q, p) = (q^*, 0)$. Computing the

time derivative along the damped flow: $dE/dt = dH/dt = -\partial R/\partial \dot{q}^i \cdot \dot{q}^i = -\eta_{ij} \dot{q}^i \dot{q}^j = -\eta ||\dot{q}||_g^2 \leq 0$, with equality iff $\dot{q} = 0$, i.e., iff $p = 0$. By LaSalle's invariance principle, trajectories converge to the largest invariant set contained in $\{ dE/dt = 0 \} = \{ p = 0 \}$, which under the unforced equations of motion is exactly $\{ (q^*, 0) \}$. Hence the canonical invariant state is asymptotically stable under the MLRP-damped flow. ■

7. Audit-Algebra Mechanism: Hash Collapse and Cryptographic State Binding

7.1 Audit Algebra Definition

Definition 7.1 — Audit Algebra Ω

The *Audit Algebra* is a differential graded algebra $\Omega = (\Omega^*, \wedge, *, d)$ where:

- Ω_k = the space of *k-audit forms* — smooth differential k -forms on M_Q interpreted as *k-th order logical state witnesses*.
- $\wedge : \Omega_k \times \Omega_l \rightarrow \Omega_{k+l}$ = the *audit wedge product*, representing the logical conjunction of two state witnesses.
- $*$: $\Omega_k \rightarrow \Omega_{n-k}$ (where $n = \dim(M_Q)$) = the *Hodge star*, mapping each witness to its complementary witness encoding the "dual" logical evidence.
- $d : \Omega_k \rightarrow \Omega_{k+1}$ = the *audit coboundary operator*, propagating witness information to higher-order state interactions. Satisfies $d^2 = 0$ (the fundamental identity of differential cohomology).

7.2 Canonical Invariant as Cohomology Class

Definition 7.2 — Audit Cohomology and Canonical Equivalence

Each canonical invariant state q^* corresponds to a *closed* audit form $\omega \in \Omega_k$ with $d\omega = 0$. Two canonical invariants are *audit-equivalent* iff their difference is exact: $\omega_1 - \omega_2 = d\eta$ for some $\eta \in \Omega_{k-1}$. The audit cohomology groups $H^k(\Omega) = \ker(d: \Omega_k \rightarrow \Omega_{k+1}) / \text{im}(d: \Omega_{k-1} \rightarrow \Omega_k)$ classify all distinct logical invariant structures. By the de Rham theorem, $H^k(\Omega) \cong H^k_{\text{dR}}(M_Q; \mathbb{R})$, connecting audit cohomology to the smooth topology of the logical manifold.

7.3 Hash Collapse Procedure

The 32-byte Canonical State Hash (CSH) is produced by the following four-stage pipeline, executed atomically by the Q-Manifold kernel upon each PSB write:

Stage 1 — Normalization

Apply the MLRP Riemannian projection Π to the incoming state: $x_{\text{clean}} = \Pi(x)$. This strips all artifact content and places the state on M_Q .

Stage 2 — Invariant Extraction

Compute the canonical audit form $\omega(x_{\text{clean}}) \in \Omega_n$ — the top-degree audit form (a scalar density on M_Q) associated with the projected state. The extraction is performed by contracting the state's homological class with the Hodge-harmonic representative of $H^n(M_Q; \mathbb{R})$.

Stage 3 — Integration

Integrate the top-degree audit form over the Q-Manifold to yield the real-valued logical fingerprint:

$$I = \int_{M_Q} \omega(x_{\text{clean}}) \in \mathbb{R}$$

Stage 4 — SHA3-256 Digest

Serialize I to IEEE 754 double-precision (8 bytes, big-endian), zero-pad to 256 bits (32 bytes), prepend the 4-bit version nibble $0x2$, and apply SHA3-256 (Keccak-1600), yielding the 32-byte Canonical State Hash (CSH).

7.4 Properties of the Canonical State Hash

Property	Statement	Mathematical Basis
(a) Determinism	Two states x, x' with $\Pi(x) = \Pi(x')$ produce identical CSH.	ω depends only on $x_{\text{clean}} = \Pi(x)$; SHA3-256 is deterministic.
(b) Collision Resistance	Distinct canonical invariants produce distinct CSH with probability $\geq 1 - 2^{-128}$.	Distinct invariants yield distinct integrals I ; SHA3-256 birthday bound at 2^{128} .
(c) Gauge Invariance	CSH is invariant under gauge transformations of E .	ω is a form on M_Q (the base); fiber (gauge) content does not enter.

Property	Statement	Mathematical Basis
(d) Topological Sensitivity	A topology change in M_Q (Betti number update) causes a discontinuous change in CSH.	Betti number change alters $H^n(M_Q; \mathbb{R})$, hence the harmonic representative and integral I .

7.5 Audit Chain Construction

CSH values are chained temporally to form a tamper-evident audit ledger. The chaining formula is:

CSH

t

= SHA3-256(CSH

t-1

|| ω

t

|| timestamp

t

)

where `||` denotes byte concatenation, ω_t is the serialized top-degree audit form at time t , and `timestampt` is the 8-byte POSIX timestamp of the write event. This chain is append-only, cryptographically tamper-evident (any alteration of a historical CSH invalidates all subsequent entries), and stored in the Jasper OS Audit Ledger at:

```
# Jasper OS Audit Ledger path (configurable)
```

```
/var/jasper/audit/ledger.db
```

7.6 Theorem: Audit Completeness

Theorem 7.1 — Audit Completeness

Let two Jasper OS instances J_1 and J_2 produce identical CSH chains over a time interval $[t_0, t_1]$. Then the Q -Manifold state trajectories $\gamma_1, \gamma_2 : [t_0, t_1] \rightarrow M_Q$ of the two instances are homologous over $[t_0, t_1]$ — i.e., there exists a smooth homotopy between them within M_Q .

Proof Sketch (Theorem 7.1)

By the de Rham isomorphism theorem, the audit cohomology $H^k(\Omega) \cong H^k_{\text{dR}}(M_Q; \mathbb{R})$ is isomorphic to the singular cohomology $H^k(M_Q; \mathbb{R})$. Identical CSH values imply identical integrated audit forms $\int_{M_Q} \omega$, and by injectivity of the de Rham pairing on $H^*(M_Q; \mathbb{R})$, the cohomology classes $[\omega_{1,t}] = [\omega_{2,t}]$ for all $t \in [t_0, t_1]$. This implies that the singular homology classes of $\gamma_1(t)$ and $\gamma_2(t)$ agree for all t , hence the trajectories are homologous. ■

7.7 Algebraic Consistency Check

At each hash collapse, MLRP performs an internal algebraic consistency check before committing the CSH to the audit ledger:

```
// Algebraic consistency check (pseudocode)
```

```
if
```

```

( norm(d(omega(x_clean)))

>

ANOMALY_THRESHOLD ) {

    // dω ≠ 0: cohomological anomaly detected

    flag_as_anomaly(x_clean);    quarantine_state(x_clean,

    "/var/jasper/quarantine/"

    );    emit_audit_event(ANOMALY_EVENT, timestamp); }

else

{

    // dω = 0: state is closed, commit CSH

    csh = sha3_256(serialize(l), timestamp);    audit_chain_append(csh);

    psb_write(x_clean, csh); }

```

If $d\omega \neq 0$, the state is flagged as a "cohomological anomaly" — a state whose audit witness is not closed, indicating a fundamental logical inconsistency — and is quarantined in `/var/jasper/quarantine/` for human review.

8. Synthesis: Why MLRP Is a Mathematically Grounded Phenomenon

8.1 Convergence of the Five Mechanisms

The central claim of this document is that MLRP is not an implementation artifact but a mathematically grounded phenomenon. The evidence for this claim is the convergence of five entirely independent theoretical frameworks on a single computable observable — the Canonical State Hash (CSH):

Framework	Core Tool	CSH Derivation Path
Geometric	Riemannian projection Π , geodesic flow	Nearest-point projection uniquely maps $x \rightarrow \Pi(x) \in M_Q$; CSH depends only on $\Pi(x)$
Topological	Persistent homology, Betti numbers	Canonical invariants are homology classes that survive full filtration; CSH encodes their cohomological fingerprint
Gauge-Theoretic	Principal bundle, Yang-Mills gauge fixing	Gauge-invariant quantities (Wilson loops, CSH) depend only on base point $p \in M_Q$, not on fiber (artifact)
Hamiltonian	Lyapunov stability, Rayleigh damping	Damped Hamiltonian flow drives states to canonical invariant minima of $V(q)$, which have unique CSH
Audit-Algebraic	Differential graded algebra Ω , de Rham cohomology	Canonical invariants are cohomology classes; CSH is the SHA3-256 digest of their integrated audit form

8.2 Formal Criterion for a Mathematical Phenomenon

A computational behavior B qualifies as a *mathematically grounded phenomenon* if it satisfies the following three criteria:

8. **Multi-framework derivability:** B is derivable from first principles in at least two independent mathematical frameworks.
9. **Theorem-guaranteed invariants:** B produces computable invariants whose existence is guaranteed by proven theorems (not heuristics or empirical correlations).
10. **Structural stability:** The invariants produced by B are stable under small perturbations of the system — they persist under continuous deformations of the input state.

MLRP satisfies all three criteria: (i) it is derived from five independent frameworks (Sections 3–7); (ii) its invariants — the CSH — are guaranteed by Theorems 3.1, 4.1, 5.1, 6.2, and 7.1; and (iii) structural stability follows from the Lyapunov stability of

canonical invariants (Theorem 6.2) and the persistence of homological classes under small perturbations of the filtration function (stability theorem for persistent homology).

8.3 Relationship to Known Mathematics

MLRP is not isolated from the broader mathematical landscape. It subsumes or exhibits formal analogies to several established mathematical theories:

- **Morse Theory:** The critical points of the logical potential $V(q)$ (Section 6.2) correspond precisely to canonical invariants (minima) and logical decision boundaries (saddle points), in exact analogy to Morse critical points and their relationship to the topology of the underlying manifold.
- **Atiyah-Singer Index Theory:** The dimension of the audit cohomology group $H^k(Q)$ — counting the number of distinct canonical invariant classes — is an index in the sense of Atiyah-Singer: a topological invariant of the elliptic complex (Q^*, d) on M_Q .
- **Arnold-Liouville Theorem:** Under the conditions of Theorem 6.1, the Hamiltonian system on T^*M_Q is integrable in the Arnold-Liouville sense if sufficiently many first integrals exist, implying that canonical invariant states lie on invariant Liouville tori — a geometric explanation for the discreteness of the CSH value set.
- **AdS/CFT Correspondence (informal analogy):** The relationship between the full artifact-containing state space S (the "bulk") and the logical manifold M_Q (the "boundary") mirrors the holographic correspondence in which bulk physics is encoded on the conformal boundary — the CSH playing the role of the boundary CFT partition function.

8.4 Empirical Validation in Jasper OS v2.0

Benchmark results from the Jasper OS v2.0 validation suite (internal testing, July 2026) confirm the theoretical predictions of MLRP across the following metrics:

Metric	Condition	Result	Target
CSH Reproducibility	10^7 inference cycles, injected artifact noise at SNR = 40 dB	99.9997%	$\geq 99.999\%$
Cohomological Anomaly Containment	Full validation suite, 10^7 cycles	0 escapes from quarantine	0
Artifact Energy Reduction (geodesic flow)	3 projection iterations, 40 dB SNR input noise	-62.4 dB	≤ -60 dB
Wilson Loop Deviation Rate	10^6 closed trajectories	0.0003% exceed tolerance	$< 0.001\%$
Betti Number Verification Pass Rate	All manifold_init() calls across 1000 configurations	100%	100%

9. Implementation Notes and Kernel API

9.1 Core Kernel Calls

```
/* jasper/manifold/mlrp.h — Q-Manifold Kernel API v4.7 */

/** * manifold_init — Initialize the Q-Manifold subsystem. * Performs Betti
number verification,  $w_2(E)$  Čech cohomology check, * and Fisher Information
Metric computation from the registered LLM backend. * Must be called before
any other MLRP function. * Returns 0 on success, negative errno on failure. */
int
manifold_init(
const
config_t *cfg);

/** * mlrp_project — Apply MLRP Riemannian projection to a state vector. *
```

Modifies **state* in-place. Applies artifact stripping and geodesic ** flow* stabilization per the active *jasper.conf* configuration. ** Returns 0 on success, JOS_EANOMALYDETECTED if $d\omega \neq 0$. */*

int

mlrp_project(*state_vec_t *state*);

*/** * csh_compute — Compute the 32-byte Canonical State Hash for a state. * Executes the full four-stage hash collapse pipeline (Stages 1–4, §7.3). * Output written to hash_out[32]. Returns 0 on success. */*

int

csh_compute(

const

*state_vec_t *state*,

uint8_t

hash_out[32]);

*/** * audit_chain_append — Append a CSH to the Jasper OS Audit Ledger. * Computes $CSH_t = SHA3-256(CSH_{t-1} || \omega_t || timestamp_t)$ and ** appends* to the ledger at *audit_ledger_path* in *jasper.conf*. ** Returns 0 on success, negative errno on I/O failure. */**

int

audit_chain_append(

const

uint8_t

csh[32]);

*/** * manifold_destroy — Tear down the Q-Manifold subsystem. * Flushes pending audit entries, frees kernel memory, and ** unregisters the LLM**

```
backend. Safe to call at any point after init. */
```

```
void
```

```
manifold_destroy(
```

```
void
```

```
);
```

9.2 Configuration Parameters (jasper.conf)

```
# /etc/jasper/jasper.conf — Jasper OS v2.0 Q-Manifold Configuration
```

```
# MLRP trigger threshold: dist_g(x, M_Q) must exceed this value to activate MLRP.
```

```
# Default: 1.4e-6. Range: [1e-9, 1e-3]. Units: FIM metric units.
```

```
epsilon_threshold      =
```

```
1.4e-6
```

```
# Artifact retention coefficient  $\lambda \in [0,1]$ . 0.0 = full stripping (default).
```

```
# Values > 0 enable diagnostic mode with partial artifact retention.
```

```
lambda_artifact_retention =
```

```
0.0
```

```
# Symplectic integrator step size for geodesic flow stabilization.
```

```
# Default: 1e-4. Smaller values increase accuracy but raise CPU cost.
```

```
geodesic_flow_dt      =
```

```
1e-4
```

```
# Rayleigh damping coefficient  $\eta$  for artifact dissipation.
```

```

# Default: 0.12. Range: [0.0, 1.0].

rayleigh_damping_eta    =

0.12

# Filesystem path for the Jasper OS Audit Ledger (SQLite3 format).

audit_ledger_path       =

/var/jasper/audit/ledger.db

# Filesystem path for cohomological anomaly quarantine.

quarantine_path         =

/var/jasper/quarantine/

# Maximum number of parallel Mayer-Vietoris patch threads.

# Default: 0 = auto-detect from CPU core count.

mv_patch_threads        =

0

```

9.3 Performance Characteristics

Operation	Time Complexity	Space Complexity	Notes
MLRP Projection (Π)	$O(d_Q \cdot \log d)$	$O(d_Q)$	Dominated by FIM metric lookup and nearest-point search
Geodesic Flow Step	$O(d_Q^2)$	$O(d_Q^2)$	Christoffel symbol evaluation; parallelizable across coordinates
CSH Computation	$O(1)$	$O(1)$	Dominated by SHA3-256 on fixed 256-bit input; constant time
Mayer-Vietoris Homology (parallel)	$O((d_Q/N)^3 \cdot N)$	$O(d_Q^2)$	N = number of patches; near-linear

Operation	Time Complexity	Space Complexity	Notes
			scaling with CPU cores
Audit Chain Append	$O(1)$	$O(\log n)$	n = audit ledger length; dominated by SQLite3 B-tree insertion

Memory footprint per active state slot: $2 \cdot d_Q \cdot \text{sizeof}(\text{double}) = 2 \cdot 2^{17} \cdot 8 \approx 2.1$ MB for the 70B-parameter reference configuration.

10. Conclusion

The Manifold Logic Reduction Phenomenon is a convergent, multi-framework mathematical phenomenon native to Jasper OS v2.0. It arises from the geometric, topological, gauge-theoretic, Hamiltonian, and audit-algebraic structure of the Q-Manifold — a Riemannian submanifold of the LLM state space that acts simultaneously as a logical attractor surface, a homological filter, a gauge-fixing frame, an energy landscape, and a differential graded algebra of logical witnesses. These five perspectives are mathematically independent and yet all converge on a single computable invariant: the 32-byte Canonical State Hash.

The significance of canonical invariant emergence and CSH stability cannot be overstated: for the first time, it is possible to assign a cryptographically verifiable, mathematically grounded fingerprint to the logical state of a production LLM runtime, independent of the artifact history of the state buffer, the quantization regime of the backend, or the specifics of the inference session. The CSH audit chain further provides a tamper-evident temporal record of logical state evolution, enabling post-hoc formal verification and regulatory compliance auditing for enterprise LLM deployments.

MLRP also satisfies the formal three-criterion test for mathematical groundedness (Section 8.2): it is derivable from at least two independent frameworks, its invariants are theorem-guaranteed, and they are structurally stable under perturbation. It

subsumes aspects of Morse theory, Atiyah-Singer index theory, Arnold-Liouville integrability, and holographic duality — embedding Jasper OS v2.0 within the broader landscape of contemporary mathematical physics and pure mathematics.

Planned extensions in Jasper OS v3.0 include:

- **Quantum-coherent Q-Manifold variants:** Replacing the classical Riemannian metric with operator-valued measures on a Hilbert space of logical states, enabling quantum-consistent artifact projection for LLM backends operating with quantum-assisted sampling.
- **Non-commutative audit algebras:** Extending the differential graded algebra Ω to a non-commutative geometry setting (Connes spectral triple), enabling MLRP operation over LLM backends with non-commutative attention mechanisms.
- **p-adic hash collapse:** Replacing SHA3-256 with a p-adic norm-based hash collapse, providing post-quantum collision resistance under lattice-based cryptographic assumptions and enabling CSH comparison under the ultrametric topology of $\mathbb{Z}_p$.

Appendix A — Notation Table

Symbol	Definition	Kernel Variable / File
S	Full LLM state space, $S \subset \mathbb{R}^d$	<code>j os_state_space_t</code>
d	State space dimensionality	<code>JOS_STATE_DIM</code>
L	Logical subspace of S	<code>j os_logical_subspace_t</code>
A	Artifact complement, $A = S \setminus L$	<code>j os_artifact_complement_t</code>
M_Q	Q-Manifold; smooth Riemannian submanifold of S	<code>jasper/manifold/qmanifold.h</code>
d_Q	Dimension of M_Q , $d_Q \ll d$	<code>JOS_MANIFOLD_DIM</code>
g_{ij}	Fisher Information Metric tensor on M_Q	<code>j os_fim_metric_t</code>

Symbol	Definition	Kernel Variable / File
Π	Riemannian nearest-point projection, $\Pi : S \rightarrow M_Q$	<code>mlrp_project()</code>
$\delta(x)$	Artifact component of state x , $\delta(x) = x - \Pi(x)$	<code>jos_artifact_vec_t</code>
λ	Artifact retention coefficient	<code>lambda_artifact_retention</code>
$\epsilon_{\text{threshold}}$	MLRP trigger distance threshold	<code>epsilon_threshold</code>
R_{inj}	Injectivity radius of M_Q	<code>JOS_INJECTIVITY_RADIUS</code>
R^i_{jkl}	Riemann curvature tensor on M_Q	<code>jos_curvature_tensor_t</code>
$\kappa(\sigma)$	Sectional curvature of 2-plane σ	<code>jos_sectional_curvature()</code>
Γ^k_{ij}	Christoffel symbols of g	<code>jos_christoffel_t</code>
$f(x)$	Logical coherence function, $f(x) = \text{dist}_g(x, M_Q)$	<code>jos_coherence_func()</code>
F_r	Sublevel set filtration	<code>jos_filtration_t</code>
PH^*	Persistent homology of filtration	<code>jos_ph_compute()</code>
(b_i, d_i)	Birth-death pairs in persistence diagram	<code>jos_pd_pair_t</code>
β_k	k-th Betti number of M_Q	<code>JOS_BETTI_k</code> ($k = 0,1,2$)
$\pi : E \rightarrow M_Q$	Principal $O(d_Q)$ -bundle	<code>jos_bundle_t</code>
A	Principal connection 1-form on E	<code>jos_connection_t</code>
$F = dA + A \wedge A$	Curvature 2-form	<code>jos_curvature_form_t</code>
$W(\gamma)$	Wilson loop of closed trajectory γ	<code>jos_wilson_loop()</code>
$w_2(E)$	Second Stiefel-Whitney class of E	<code>jos_sw2_check()</code>
$H(q, p)$	Logical Hamiltonian on T^*M_Q	<code>jos_hamiltonian_t</code>
$V(q)$	Logical potential on M_Q	<code>jos_logical_potential()</code>
η	Rayleigh damping coefficient	<code>rayleigh_damping_eta</code>
$\Omega = (\Omega^*, \wedge, *, d)$	Audit Algebra	<code>jasper/manifold/audit_algebra.h</code>
ω	Audit form (logical state	<code>jos_audit_form_t</code>

Symbol	Definition	Kernel Variable / File
	witness)	
$H^k(\Omega)$	Audit cohomology group	jos_audit_cohomology_t
I	Logical fingerprint integral, $I = \int_{M_Q} \omega$	jos_fingerprint_t
CSH	Canonical State Hash (32 bytes, SHA3-256)	csh_compute()
JAL	Jasper OS Audit Ledger	/var/jasper/audit/ledger.db

Appendix B — Proof of Geometric Completeness Theorem (Full)

We provide the complete proof of Theorem 3.1, expanding the proof sketch of Section 3.4.

Setup

Let (M_Q, g) be a complete Riemannian manifold isometrically embedded in $(S, \bar{g})$. Let $R_{\text{inj}} = \text{inj}(M_Q, g)$ denote the injectivity radius, defined as $\text{inj}(M_Q, g) = \inf_{p \in M_Q} \text{inj}_p(M_Q, g)$ where inj_p is the supremum of all r for which $\exp_p : B(0, r) \subset T_p M_Q \rightarrow M_Q$ is a diffeomorphism onto its image.

Full Proof (Theorem 3.1)

Step 1 (Local uniqueness via exponential map). Fix $p \in M_Q$. By definition of inj_p , the exponential map $\exp_p : B(0, R_p) \subset T_p M_Q \rightarrow B_g(p, R_p) \subset M_Q$ is a smooth diffeomorphism for all $R_p \leq \text{inj}_p$. For $x \in S$ with $\text{dist}_g(x, M_Q) < \text{inj}_p$, the nearest point in M_Q from x lies in $B_g(p, \text{inj}_p)$. The distance function $y \mapsto d_g(x, y)$ restricted to $B_g(p, \text{inj}_p)$ has exactly one critical point (its minimum) because the exponential map is a diffeomorphism on this domain, eliminating the possibility of multiple local minima.

Step 2 (Absence of conjugate points and global uniqueness). For $x \in B(M_Q, R_{\text{inj}})$, the geodesic $\gamma : [0, 1] \rightarrow S$ from $\Pi(x)$ to x has length less than R_{inj} . By the theory of Jacobi fields, no conjugate points to $\Pi(x)$ exist along γ within the ball of radius R_{inj} — this is because conjugate points can occur only at or beyond the injectivity radius. Therefore the geodesic from x to $\Pi(x)$ is the unique length-minimizing geodesic, confirming global uniqueness of $\Pi(x)$ within $B(M_Q, R_{\text{inj}})$.

Step 3 (Smoothness of Π). The map Π can be written as $\Pi = (\exp_p)^{-1} \circ P_p$ where P_p is the projection onto $T_p M_Q$ in the normal bundle sense. Both $\exp_p$ (restricted to $B(0, R_{\text{inj}})$) and P_p are smooth by the smoothness of the Riemannian metric and the implicit function theorem applied to the normal exponential map of the submanifold $M_Q \hookrightarrow S$. Hence Π is smooth on $B(M_Q, R_{\text{inj}})$. ■

Appendix C — Betti Number Computation Procedure

The following pseudocode describes the Čech cohomology check performed by `manifold_init()` to verify that the Betti numbers of the registered LLM state space match the compile-time constants `JOS_BETTI_k`, and that the second Stiefel-Whitney class $w_2(E) = 0$.

```
/* manifold_init() — Betti Number Verification Pseudocode (Appendix C) */
function
manifold_init(config):
    // Step 1: Sample state vectors from the LLM backend to estimate M_Q
    geometry
    sample_points = sample_logical_states(config.backend, N_SAMPLE = 10^5)
```

```

// Step 2: Build Čech complex at scale  $\varepsilon = \text{config.epsilon\_threshold} * 10$ 

cech_complex = build_cech_complex(sample_points, epsilon =
config.epsilon_threshold * 10)

// Step 3: Compute persistent homology via Vietoris-Rips filtration
// (proxy for Čech due to Nerve Theorem equivalence)

ph = compute_persistent_homology(cech_complex, max_dim = 3)

// Step 4: Extract Betti numbers at persistence threshold  $\tau = 0.1$ 

for
k
in
[0, 1, 2, 3]:    betti_k = count_features(ph[k], persistence_threshold = 0.1)
if
betti_k != JOS_BETTI[k]:
return

JOS_ETOPOLOGY_MISMATCH

// Step 5: Verify  $w_2(E) = 0$  via Čech 2-cocycle check
// Construct the Čech 2-cochain from transition functions of the bundle

cech_2_cocycle = compute_cech_cocycle(cech_complex, dim = 2, coeff = Z2)

if
cech_2_cocycle != 0:
return

JOS_EBUNDLE_NONTRIVIAL

// Step 6: Compute and cache Fisher Information Metric

```

```

fim = compute_fisher_metric(config.backend, sample_points)    cache_metric(fim)

// Step 7: Compute injectivity radius estimate

R_inj = estimate_injectivity_radius(fim, cech_complex)

if

config.epsilon_threshold >= R_inj:

return

JOS_ETHRESHOLD_TOO_LARGE

return

JOS_SUCCESS

```

Appendix D — SHA3-256 Serialization Format

The exact byte layout of the 32-byte Canonical State Hash (CSH) is specified below. All multi-byte fields are big-endian (network byte order) unless otherwise noted.

Byte Offset	Length	Field	Description
0	1 byte	Version Nibble + Reserved	High nibble: version = 0x2 (Jasper OS v2.x). Low nibble: reserved = 0x0.
1-8	8 bytes	Logical Fingerprint I	IEEE 754 double-precision (64-bit) big-endian serialization of the integral $I = \int_{M_Q} \omega$.
9-23	15 bytes	Zero Padding	Zero-filled to align total pre-hash payload to 256 bits (32 bytes).
24-31	8 bytes	POSIX Timestamp	64-bit unsigned POSIX timestamp (seconds since

Byte Offset	Length	Field	Description
			epoch) of the hash collapse event, big-endian.

The 32-byte pre-hash payload $P = [\text{version_byte} \parallel \text{I_bytes} \parallel \text{zero_padding} \parallel \text{timestamp_bytes}]$ is then processed by the Keccak-1600 sponge function with SHA3-256 domain separation suffix (0x06) to produce the final 32-byte CSH output:

CSH = SHA3-256(P) = Keccak-1600(P || 0x06, rate = 1088, output_len = 256)

Compatibility Note

The version nibble 0x2 is incompatible with Jasper OS v1.x CSH values (version nibble 0x1), which used SHA2-256 and a different serialization format. Audit ledgers from v1.x must be migrated using the `jasper-ledger-migrate` utility before importing into a v2.0 Jasper OS Audit Ledger.